

ECON 0150 | Homework 1.1

Due: Friday, Sept. 4 at 5PM

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely. You are welcomed and encouraged to work in groups so long as your work is your own. Submit your figures and answers to Gradescope.

Lets use the class survey dataset to understand who is in our class!

Q1. Data Selection

- a) Who collected this data?
- b) How was this data collected?
- c) What does this data miss?

Q2. Data Diagram

- a) What is the index variable?
- b) What is the data structure?
- c) How many (meaningful) variables does the dataset contain?

Q3. What is each of the following variable's type?

- a) "What is your favorite color?"
 - Binary Categorical
 - Nominal Categorical
 - Ordinal Categorical
 - Discrete Numerical

- Continuous Numerical

b) “Approximately how many miles away from Pittsburgh is your hometown?”

- Binary Categorical
- Nominal Categorical
- Ordinal Categorical
- Discrete Numerical
- Continuous Numerical

c) “What is your (primary) major?”

- Binary Categorical
- Nominal Categorical
- Ordinal Categorical
- Discrete Numerical
- Continuous Numerical

d) “How much did you like your statistics class?”

- Binary Categorical
- Nominal Categorical
- Ordinal Categorical
- Discrete Numerical
- Continuous Numerical

e) “How excited are you for this class?”

- Binary Categorical
- Nominal Categorical
- Ordinal Categorical
- Discrete Numerical
- Continuous Numerical

f) “Do you feel confident in Excel?”

- Binary Categorical
- Nominal Categorical
- Ordinal Categorical
- Discrete Numerical
- Continuous Numerical

Q4. Summarize ‘Do you feel confident in Excel?’.

- a) Appropriately visualize the variable ‘Do you feel confident in Excel?’.
- b) How many students in the class feel confident in Excel?

Q5. Summarize ‘What is your major?’.

- a) Appropriately visualize the variable ‘What is your major?’.
- b) How many students in the dataset are not economics majors?